TAUT CONTINUOUS HOMOMORPHISMS PRESERVE WEAK PRE-ADMISSIBILITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof span cites one same-chapter lemma, delivered as a navigation stub. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Tautness of a continuous homomorphism of linearly topologized rings. Same chapter, Definition definition-taut. Gives that the closure of IB is open for a weak ideal of definition I of A.
- The topologically nilpotent elements of a linearly topologized ring form a closed radical ideal, and the closure of the image of a set of topologically nilpotent elements consists of topologically nilpotent elements. Same chapter, Lemma lemma-topologically-nilpotent, invoked at proof lines 1430-1431. Shows that the closure of IB consists of topologically nilpotent elements, so it is a weak ideal of definition.

Source reference. Original lemma 0AMW: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1134–1134. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $\varphi : A \to B$ be a continuous homomorphism of linearly topologized rings. If φ is taut and A is weakly pre-admissible, then B is weakly pre-admissible.*

Proof. Let $I \subset A$ be a weak ideal of definition. Then the closure J of IB is open and consists of topologically nilpotent elements by Lemma 0AMW. Hence J is a weak ideal of definition of B. □

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.